**CHE210: *DISCUSSION SESSION 6*** **NAME:**______________________________

OCTOBER 8, 2019

**Evaluate the streams:**

1. 75mL $CCl_4$ (SG=1.59)

   ⟶ Calculate $\dot{n}$ (mol $CCl_4$)

2. $\dot{m}$ = 100g/h

   ⟶ Calculate $\dot{m}_{C2H4}$

   $x_{CH4} = 0.3$ ($CH_4$ mol/mol)

   $x_{C2H4} = 0.4$ ($C_2H_4$ mol/mol)

   $x_{C2H6} = 0.3$ ($C_2H_6$ mol/mol)

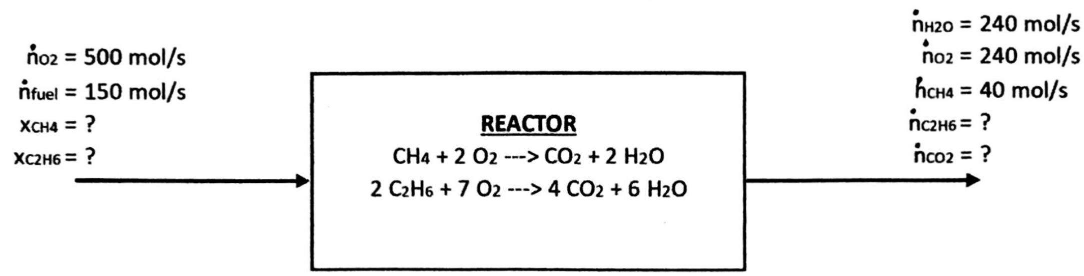
3. $\dot{n}$ = 250 kmol/h

   ⟶ Calculate $\dot{n}_{H2O}$ [in terms of x]

   $\omega_{H2O} = x$ [kg $H_2O$/kg]

   $\omega_{H2} = (1 - x)$ [kg $H_2$/kg]

4. Below is a flow digram of combustion reactor that is fed a methane/ethane fuel at an unknown mixture composition. Use the information below to determine its composition. [Use the extent of reactions!] Hint: Consider $X_{CH_4}$ and $X_{C_2H_6}$ as mol % of only the fuel.




*Degree of Freedom Analysis:*

| **Unknowns:** | | **Equations:** | |
|---|---|---|---|
| 4a. | ______ | 5a. | ______ |
| 4b. | ______ | 5b. | ______ |
| **TOTAL** | ______ | 5c. | ______ |
| | | 5d. | ______ |
| | | **TOTAL** | ______ |

*5. A drought has caused the price of strawberries to dramatically increase $4.00 per lb. The CEO of a jam company is worried that they are no longer making profits at the current $2.60 per jar of jam. Assume each jar contains approximately 1lb of jam and the current price of sugar is $0.80/lb. Use the process diagram below to determine whether the jam company is still profitable.*